



RF Project 2025

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Submitted to: Eng. Ahmed Al-Sayed

June 2025

1 Introduction

This report is submitted to Eng. Ahmed Al-Sayed. It details all the required specifications for the Low Noise Amplifier (LNA) and Mixer designed for applications operating at a frequency of 2.4 GHz.

2 Specifications

Part A: LNA Design (15 Marks)

Design an inductive-degenerated LNA at 2.4 GHz center frequency with the following specifications:

- Gain greater than 15 dB
- Noise Figure less than 2 dB
- S11 less than -10 dB

3 Simulations

The following simulations are required:

- A screenshot of your design clearly indicating the operating point, sizing, and current. (2 marks)
- Voltage gain in dB across frequency. (3 marks)
- Noise Figure in dB across frequency. (3 marks)
- S11 in dB across frequency. (3 marks)
- Output voltage versus input voltage curve using large signal simulation and measuring the IP1dB at the center frequency. (2 marks).

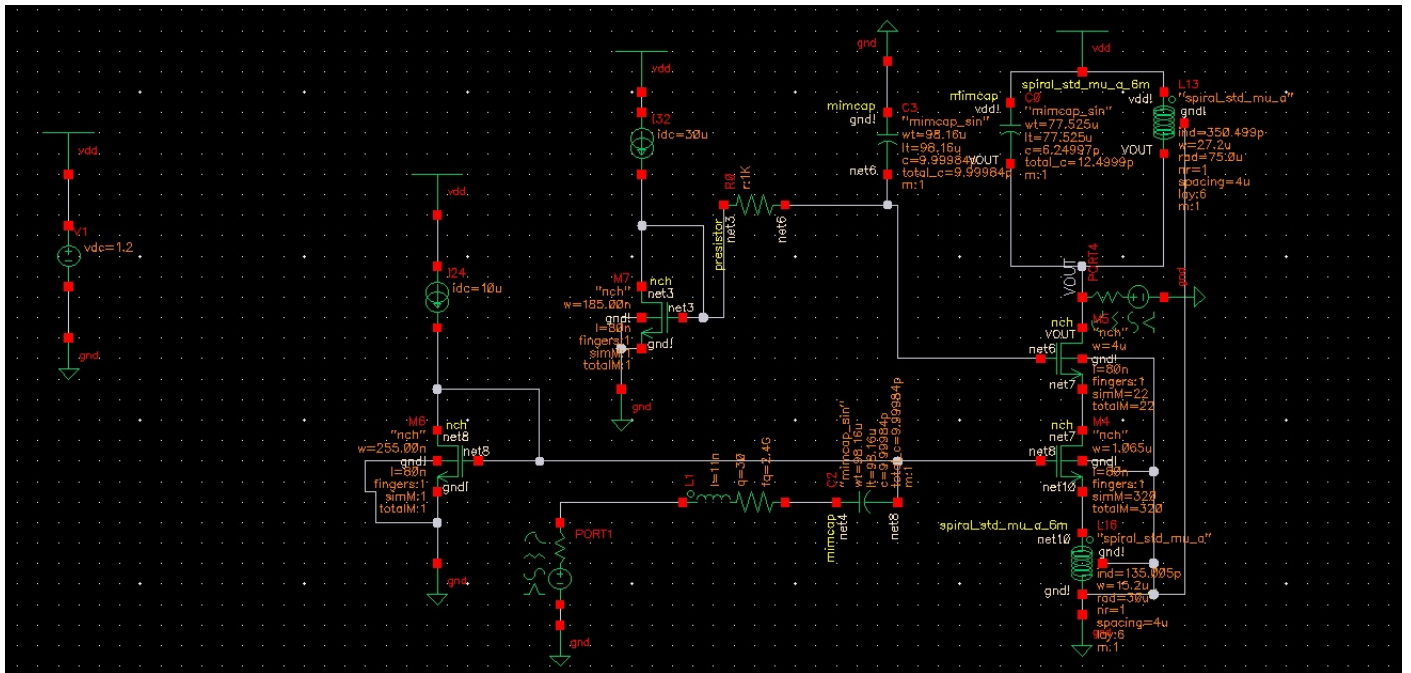


Figure 1: Schematic and sizing.

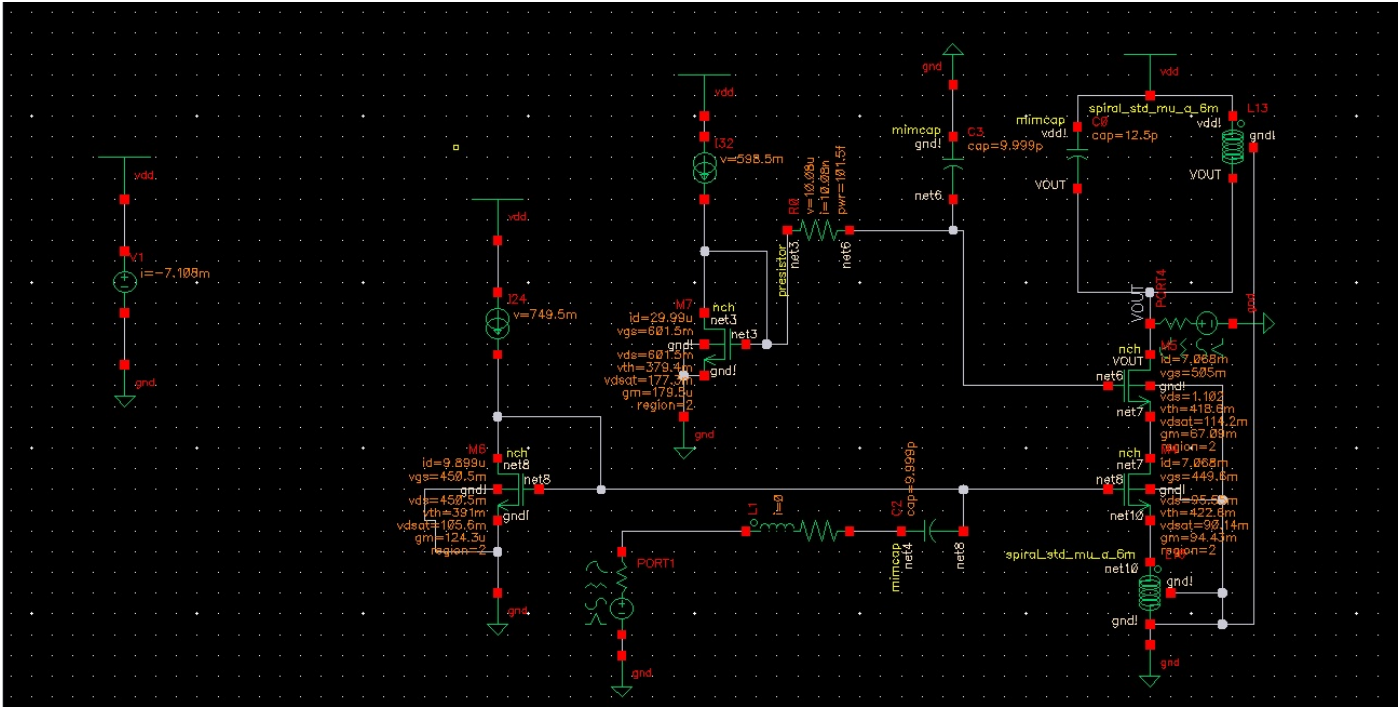


Figure 2: Design screenshot showing the operating point and current.

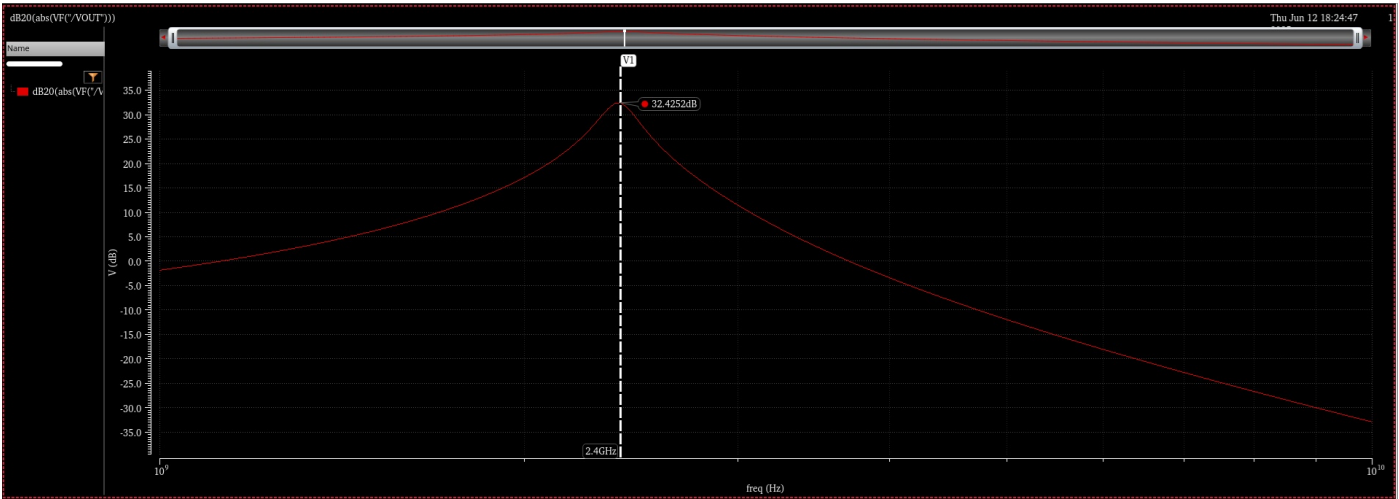


Figure 3: Voltage gain across frequency.

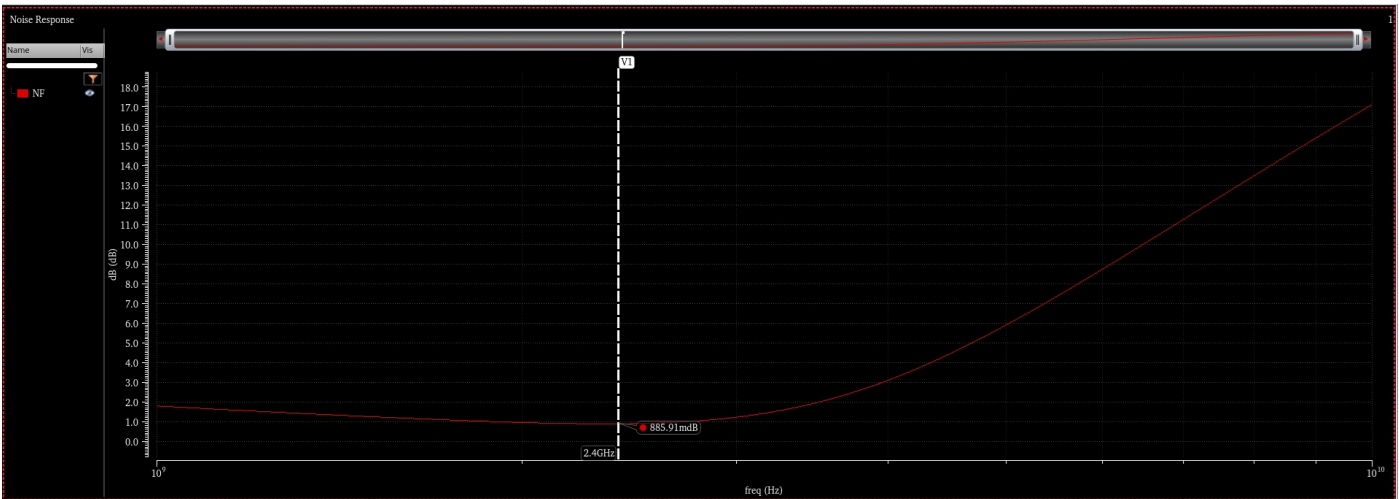


Figure 4: Noise Figure in dB across frequency.

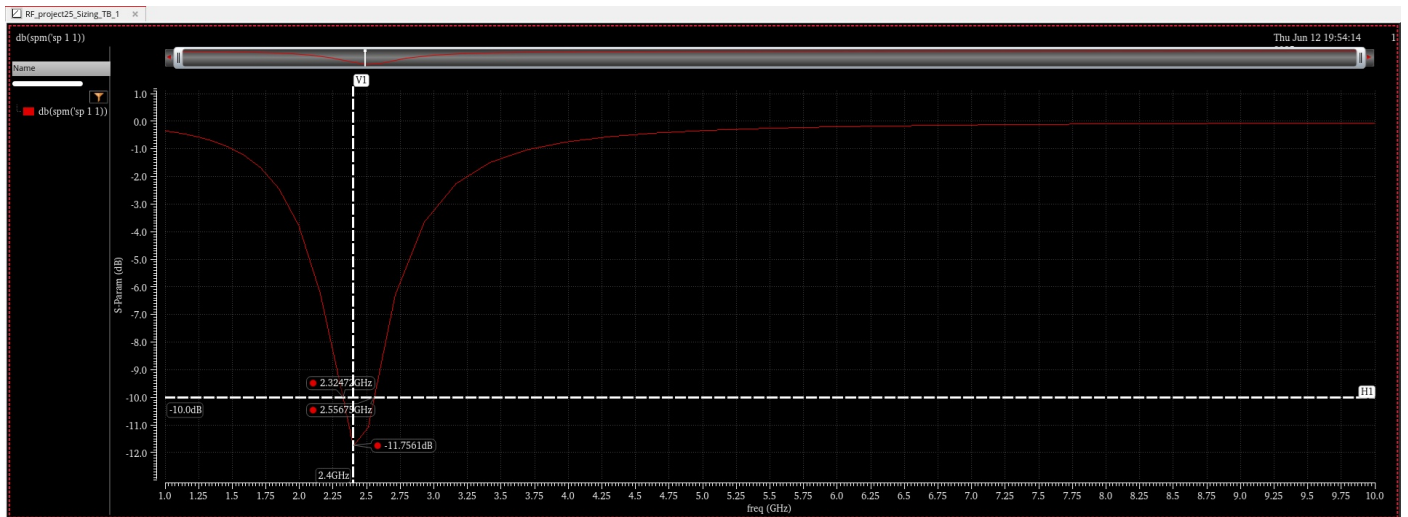


Figure 5: S11 in dB across frequency.

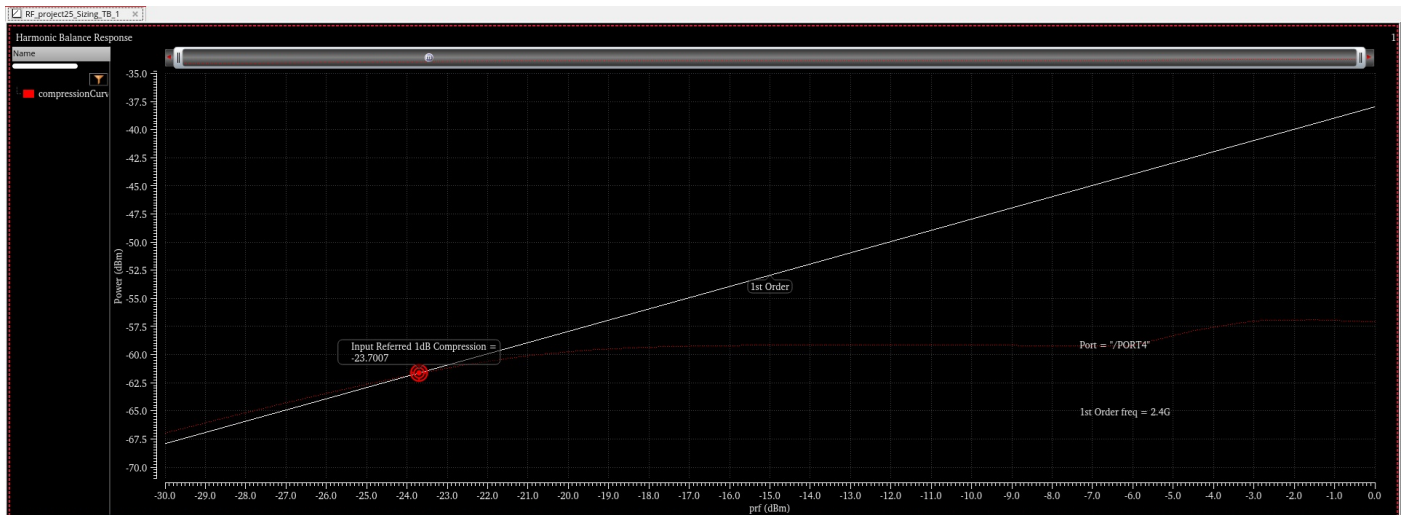


Figure 6: IP1-dB.

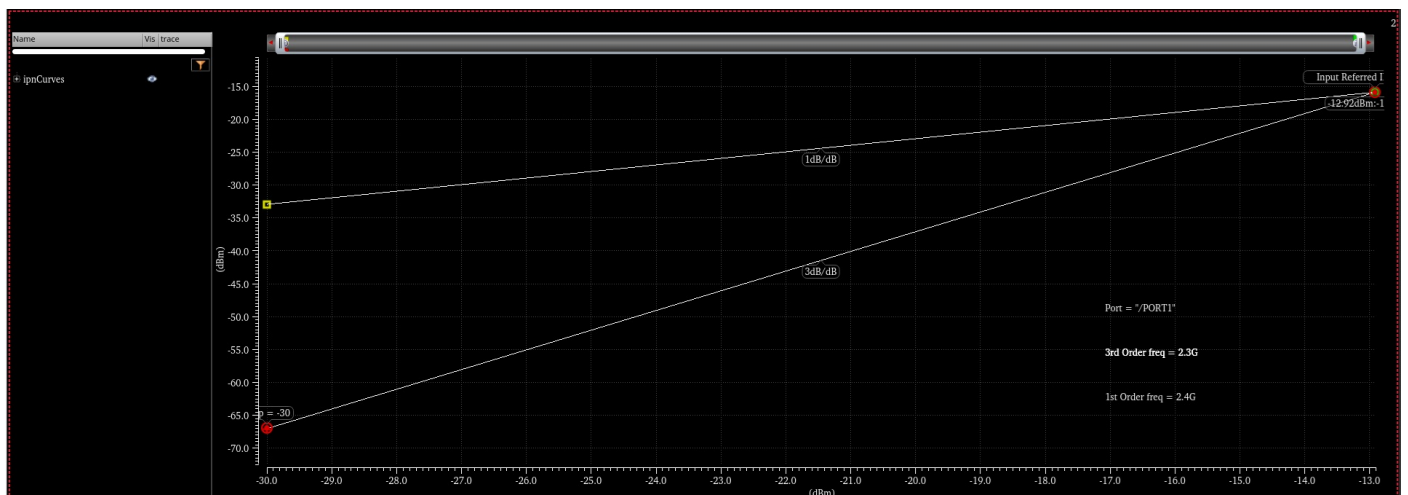


Figure 7: IIP3-dB @-30dBm Blocker.

Part B: Mixer Design

1. Mixer Overview

A double-balanced active down-conversion mixer was designed to operate at an RF frequency of 2.4 GHz, with a 30 MHz IF and 10 MHz signal bandwidth. The mixer is optimized for high conversion gain, acceptable noise figure, and good linearity, per the project specifications.

2. Design Requirements

- Conversion Gain greater than 12 dB
- DSB Noise Figure Less than 5 dB
- IIP3 greater than -8 dBm
- IP1dB greater than -18 dBm

3. LO Drive Level

The minimum required LO amplitude for proper switching of the mixer transistors was determined through parametric simulation. A value of 400mV was chosen as it ensures full current steering while minimizing LO feedthrough. Calculation: In order to have full steering current input voltage should be equal to $\sqrt{2} \cdot V_{ov}$ and we selected the value of V_{ov} to be equal to 270mV therefore LO Drive Level is roughly equal to 400mV

4. Design Schematic

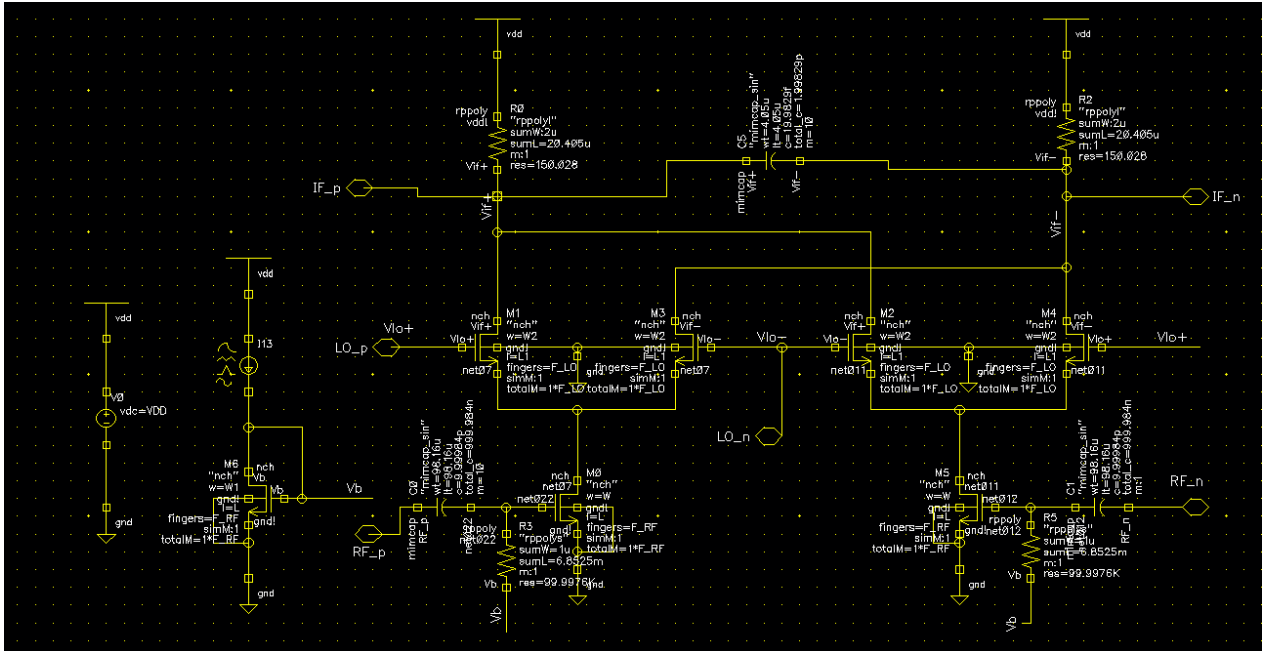


Figure 8: Mixer schematic with operating point, sizing, and current.

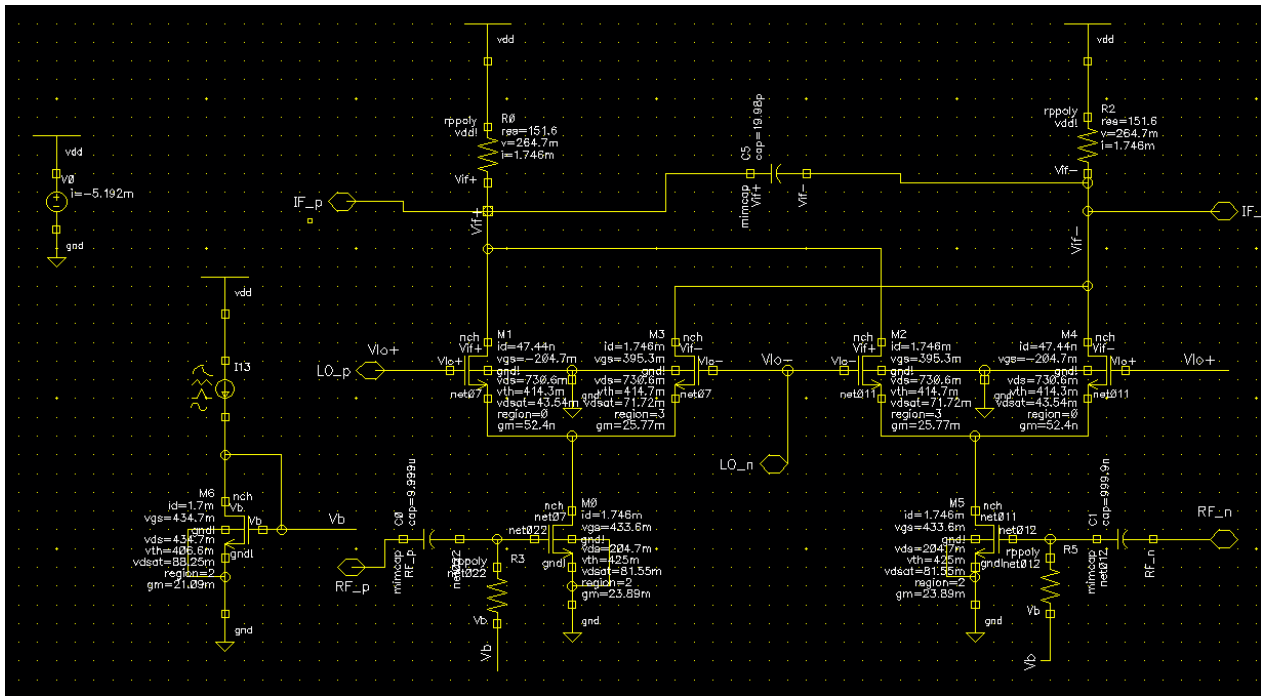


Figure 9: Mixer schematic with operating point, sizing, and current (Full steering).

5. Simulation Results

- **Conversion Gain vs LO Amplitude:** Achieves greater than 12 dB at optimal LO.
- **Noise Figure vs Frequency:** Remains below 5 dB across the band.
- **Conversion Gain vs RF Amplitude:** Used to extract IP1dB.
- **IIP3 Estimation:** IM3 and IF components were plotted and extrapolated.

Insert figures:

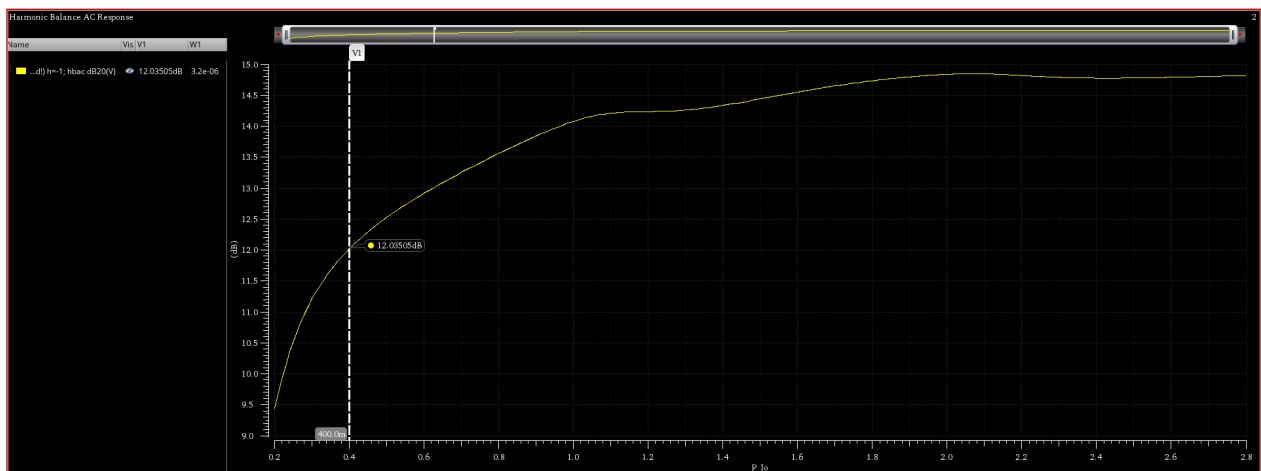


Figure 10: Conversion gain versus LO amplitude.

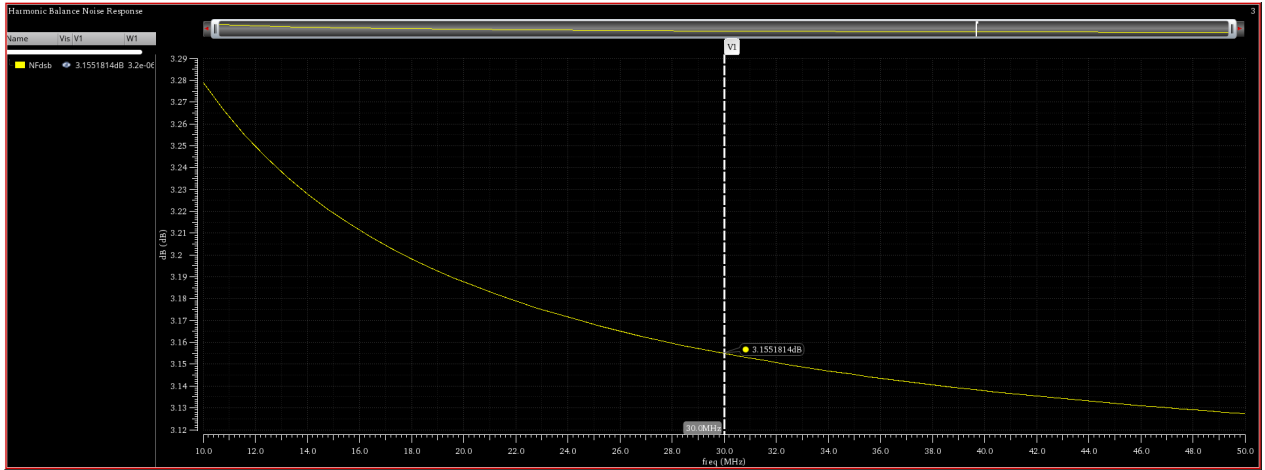


Figure 11: Mixer noise figure versus @30MHz.

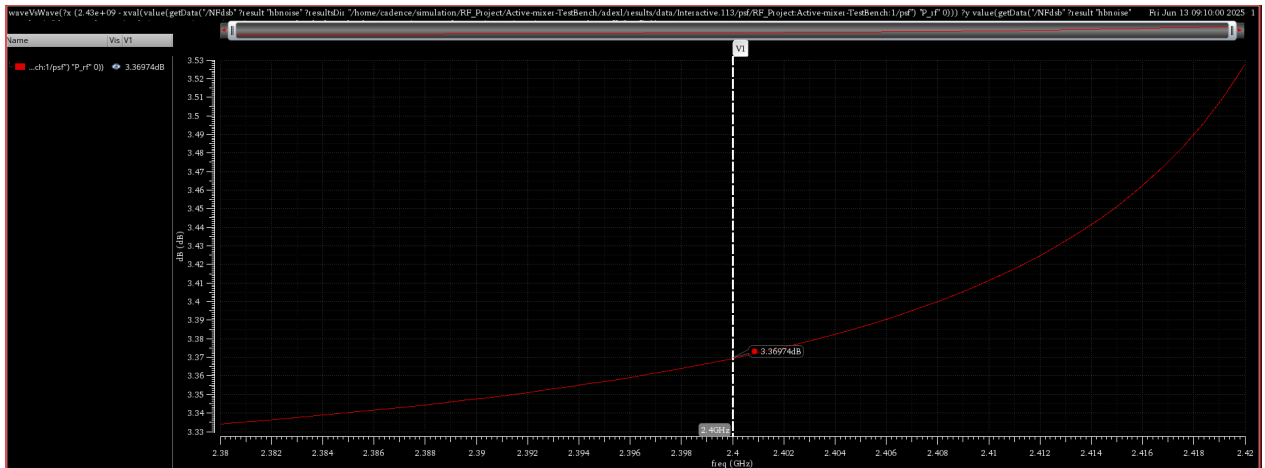


Figure 12: Mixer noise figure versus @2.4GHz.

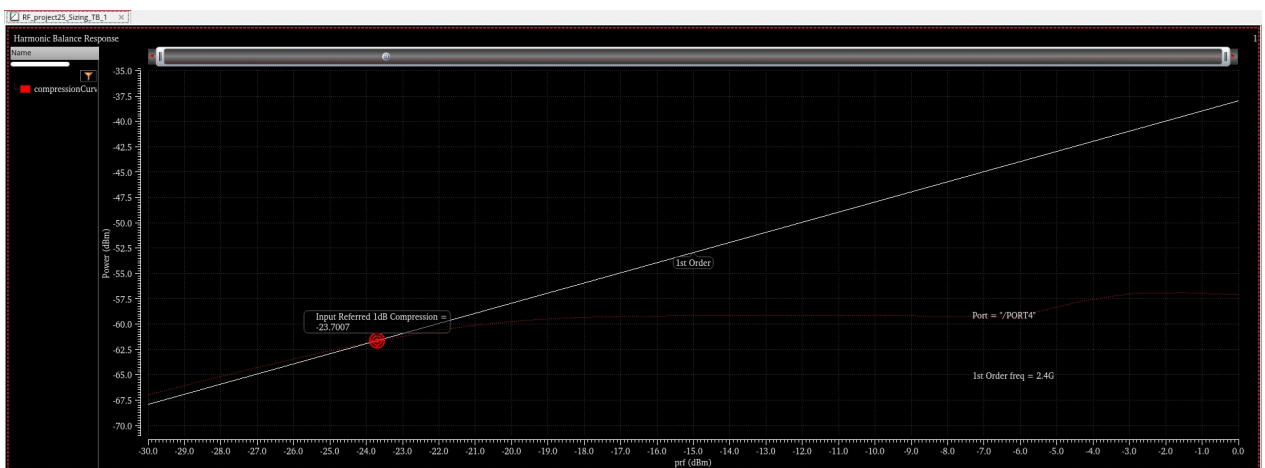


Figure 13: Conversion gain vs RF input showing IP1dB point.

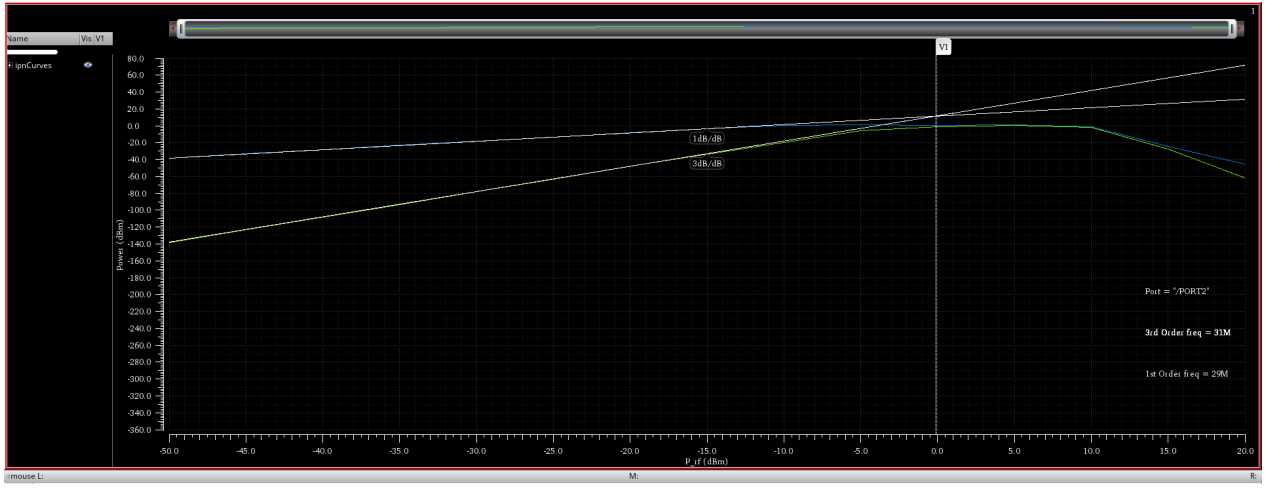


Figure 14: IM3 and IF signal levels used to extract IIP3.

Conclusion: The integrated system meets the specifications for Wi-Fi reception, with good gain, acceptable noise, and sufficient linearity. Mixer and LNA stages were designed and simulated successfully in Cadence.

Part C: RF System Integration

1. System Overview

The LNA and Mixer are cascaded to form a complete receiver front-end targeting the 2.4 GHz Wi-Fi band. System-level performance is evaluated across the relevant RF band.

2. S11 Across the Band

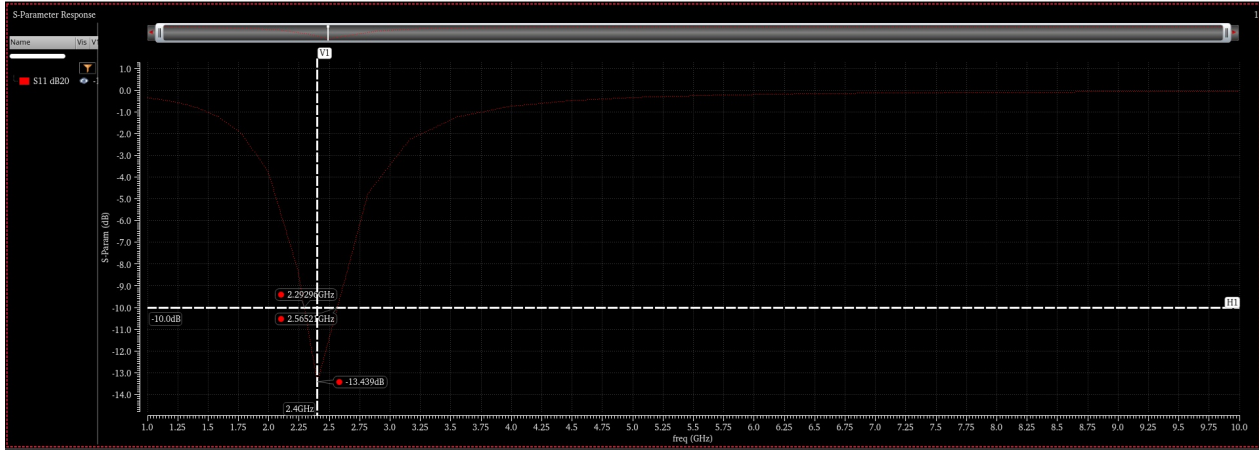


Figure 15: Input reflection coefficient (S11) across Wi-Fi band.

Achieved S11 $\leq$ -10 dB across the band, indicating good input matching.

3. Conversion Gain

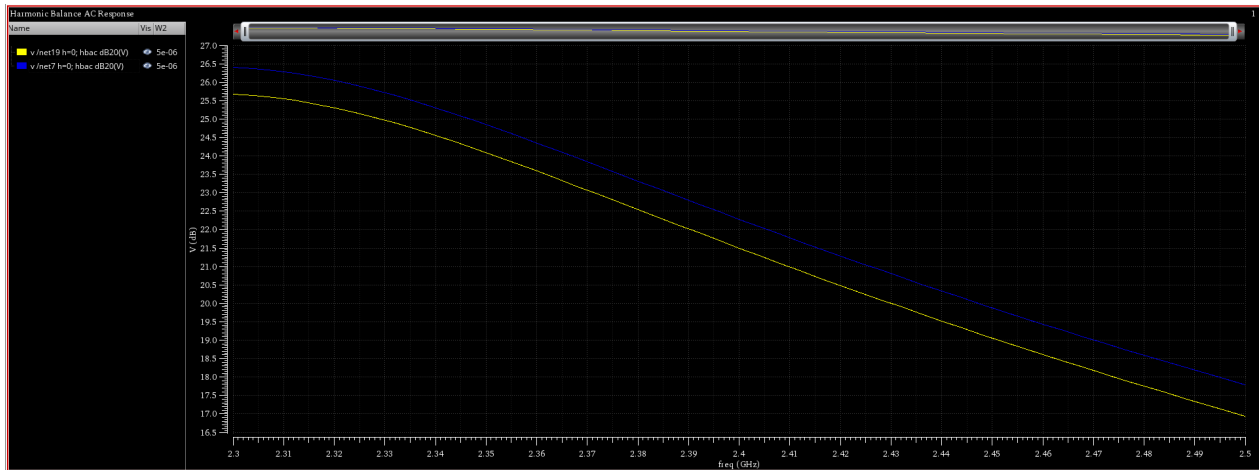


Figure 16: Gain after LNA(yellow trace), Gain after MN (Green trace).

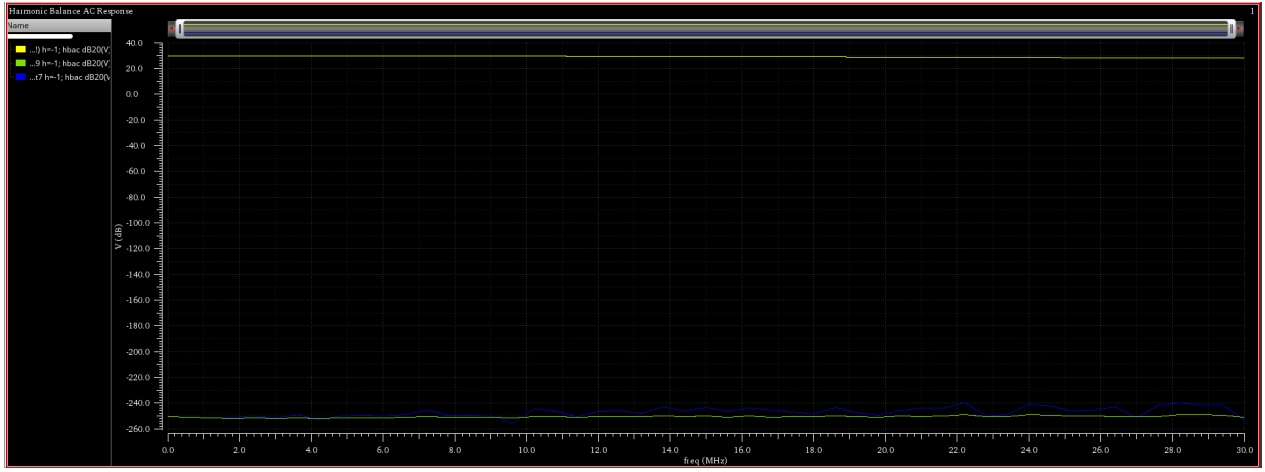


Figure 17: Gain after Mixer(yellow trace), Gain after MN (Blue trace), Gain after LNA (Green trace).

4. System Noise Figure

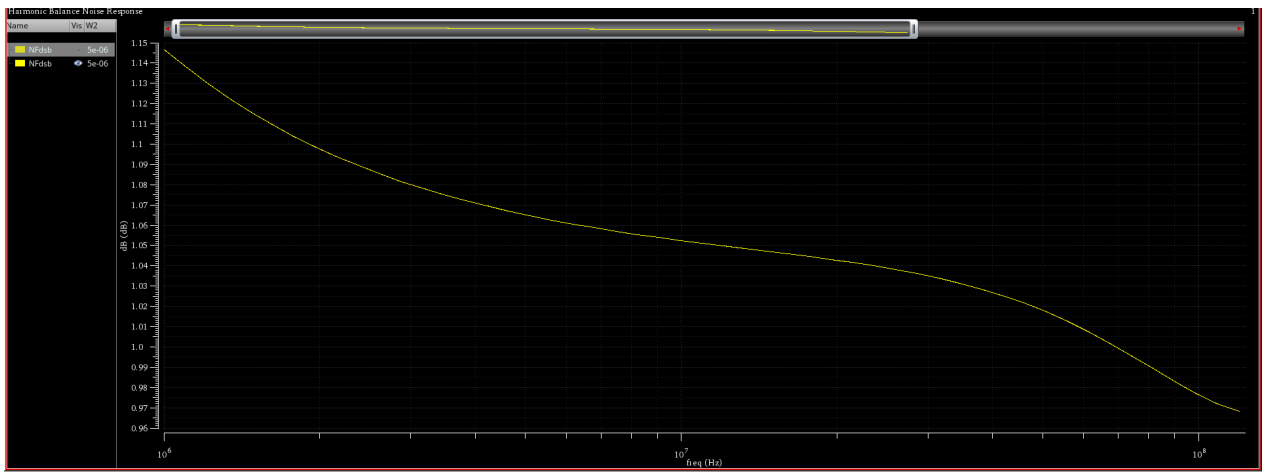


Figure 18: Simulated noise figure after LNA + Mixer.

5. IP1dB and IIP3

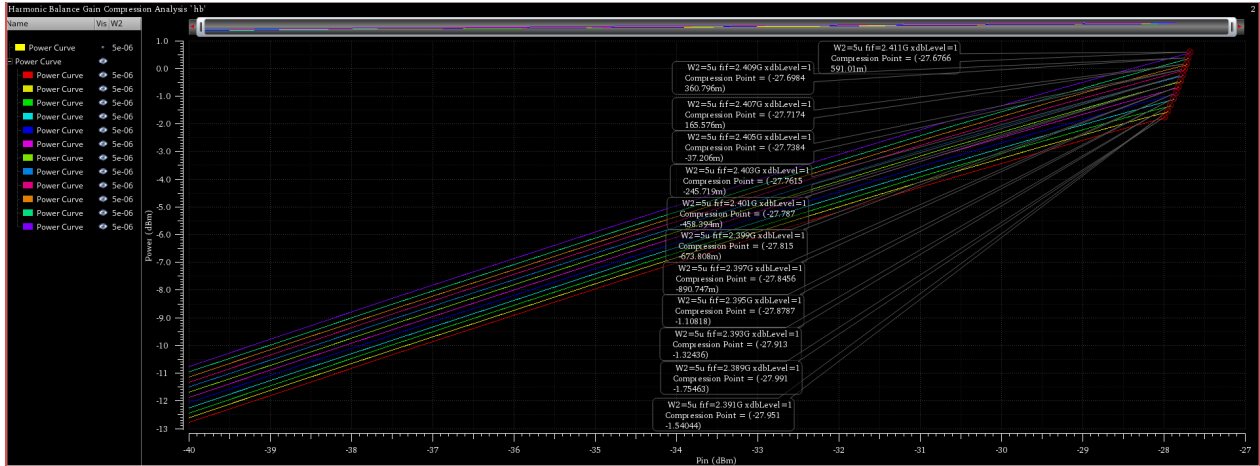


Figure 19: IP1dB of the cascaded LNA and Mixer and Sweeping the Frequency (2.391-2.411GHz).

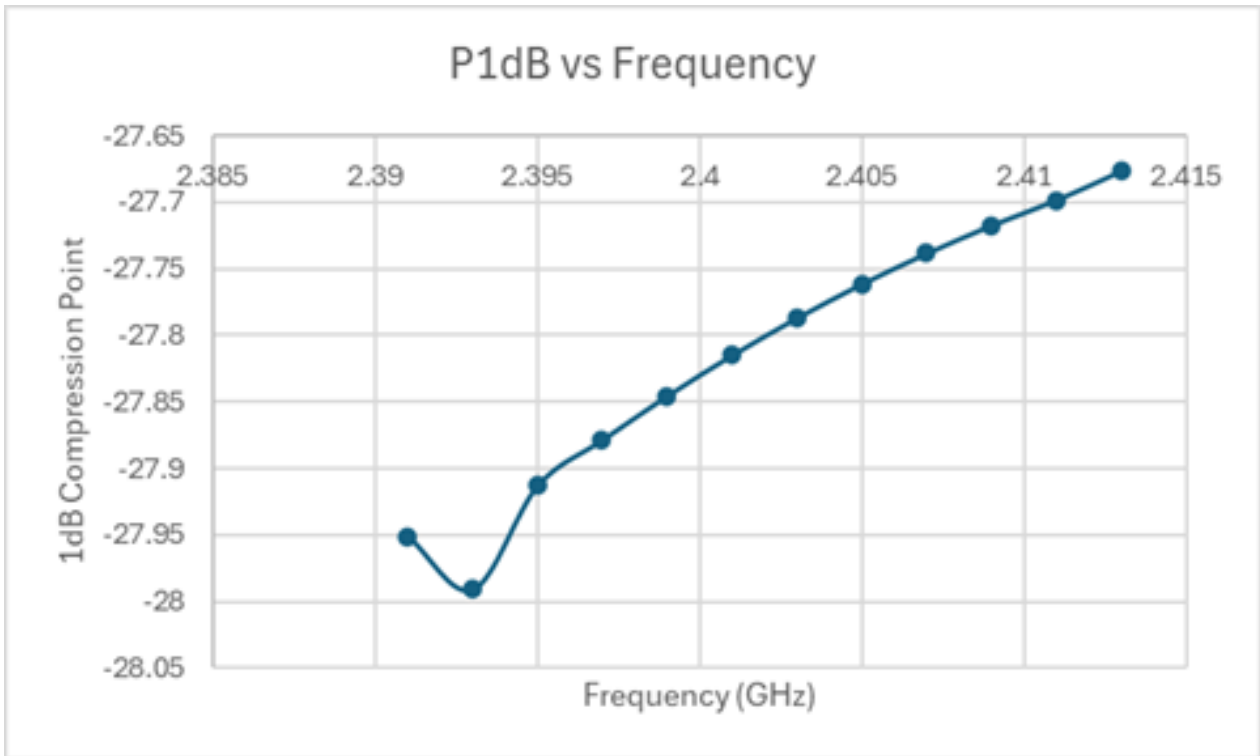


Figure 20: IP1dB of the cascaded LNA and Mixer and Sweeping the Frequency (2.38875GHz-2.41125GHz).

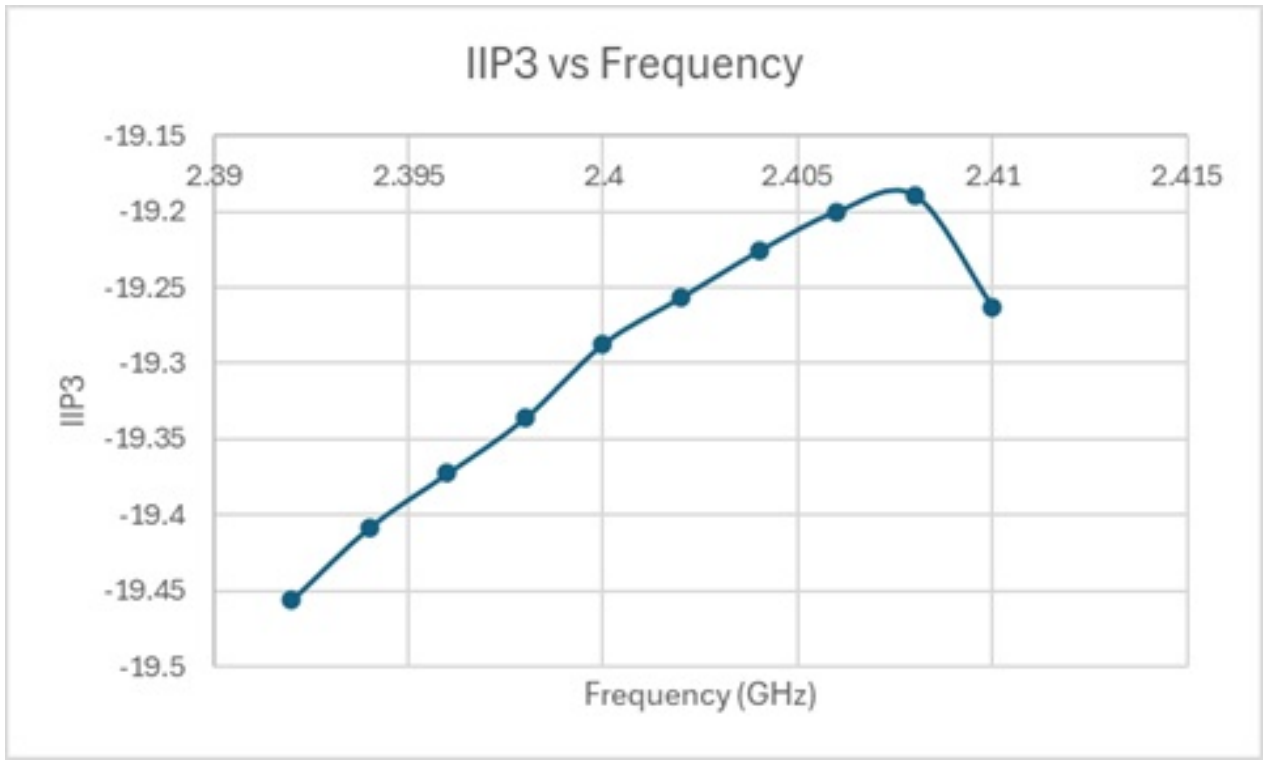


Figure 21: IIP3 of the cascaded system vs Frequency.

6. Inter-stage Matching

A matching network was evaluated between the LNA and mixer to ensure optimal power transfer. Our LNA output impedance across frequency is as shown in the figure below: The mixer input impedance is

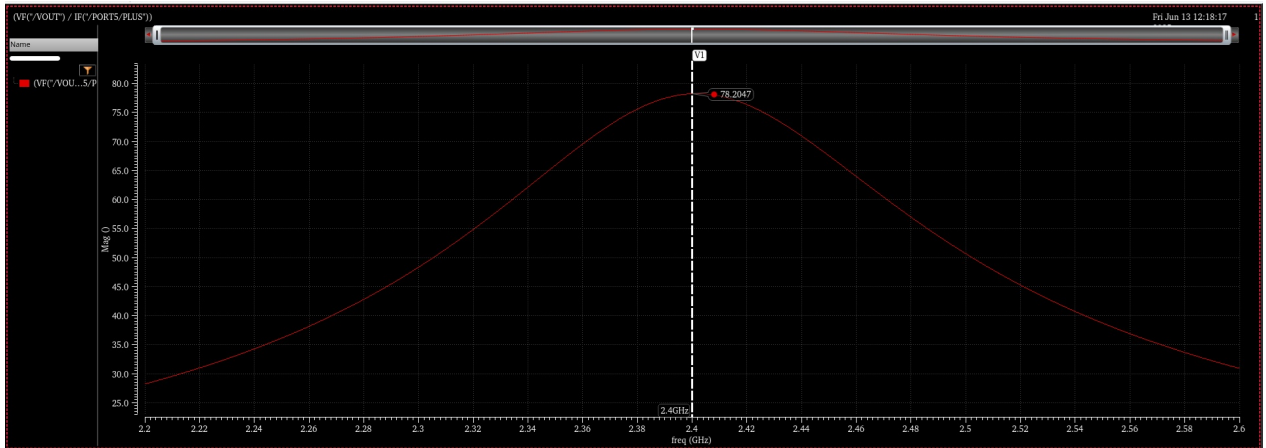


Figure 22: LNA Output Impedance Frequency .

50 ohms since we used a Balun therefore we need to perform matching Impedance. We need to match 50 ohms (Input impedance of the Mixer) to 78 ohms (Output impedance of the LNA). This is done by using a C-network, we got initial design points from ADI website and done some sweeps to get the suitable response to match impedance. Inductance equal to 3.2nH and Capacitance equal to 985fF

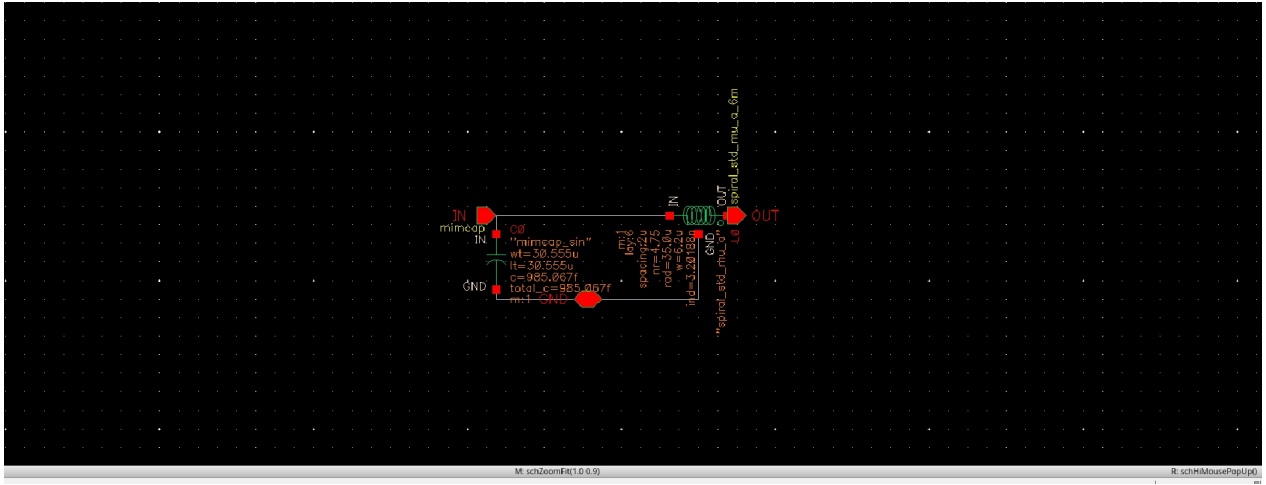


Figure 23: Matching Network Schematic.

This a response of the Matching Network verifying that it matches the output impedance.

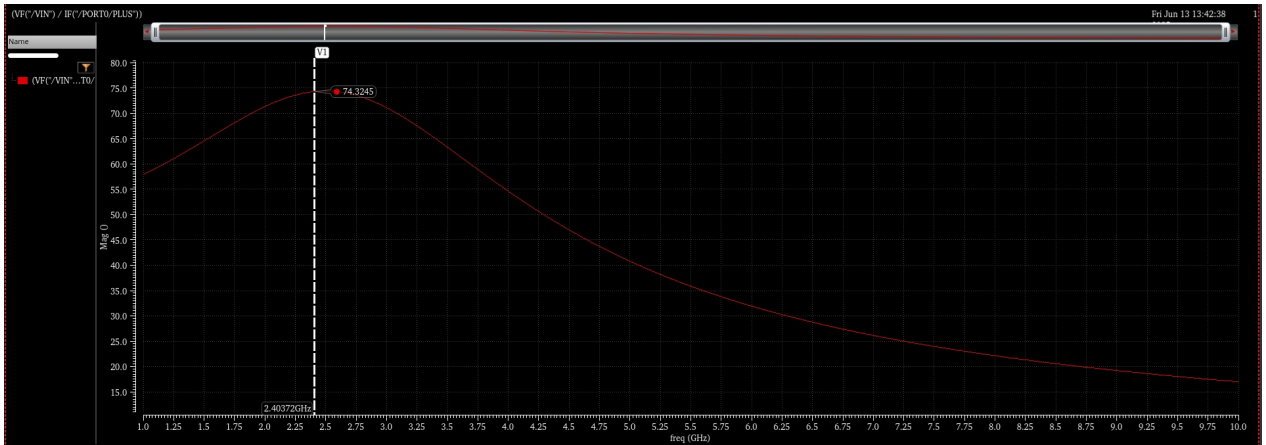


Figure 24: Matching Network Response.